higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

STRENGTH OF MATERIALS AND STRUCTURES N6

11 APRIL 2019

This marking guideline consists of 9 pages.

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QUESTION 1

1.1 $$At\ \ 70\ \ mm\ :a+\frac{b}{0,07^2}=30\times10^6\ldots\ldots\ldots(1)$$

$$at\ \ 120\ \ mm\ :a+\frac{b}{0,12^2}=0\ldots\ldots\ldots\ldots\ldots\ldots(2)$$

$$(1)-(2)\ :204,0816\,b-69,444\,b=30\times10^6$$

$$b=222,821\times10^3$$

$$a=-15,474\times10^6$$

$$at\ \ 70\ \ mm\ :\ \sigma_{Hmax}=a-\frac{b}{0,07^2}$$

$$=-15,474\times10^6-\frac{222,821\times10^3}{0,07^2}$$

$$\sigma_{Hmax}=-60,947\ \ MPa\ \ (tensile)$$

$$at\ \ 120\ \ mm\ :\ \sigma_{Hmin}=a-\frac{b}{0,12^2}$$

$$=-15,474\times10^6-\frac{222,821\times10^3}{0,12^2}$$

$$\sigma_{Hmin}=-30,947\ \ MPa\ \ (tensile)$$ (6)

1.2 $$\delta d_1=\frac{D_c}{E}(\sigma_H-\vartheta\times\sigma_R)$$

$$=\frac{0,07}{200\times10^9}(30\times10^6-0,29\times30\times10^6)$$

$$\delta d_1=7,455\times10^{-6}\,m$$ (2)

1.3 $$\delta d_2=\frac{D_c}{E}(\sigma_H-\vartheta\times\sigma_R)$$

$$=\frac{0,07}{41\times10^9}(-60,947\times10^6-0,3\times30\times10^6)$$

$$\delta d_2=-119,422\times10^{-6}\,m$$ (2)

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1.4 $\Delta d = \delta d_1 - \delta d_2$

$= 7,455 \times 10^{-6} - \left(-119,422 \times 10^{-6}\right)$

$\Delta d = 0,127 \ \ mm$ (1)

1.5 $F_\mu = \mu \times P \times \pi DL$

$= 0,2 \times 30 \times 10^6 \times \pi \times 0,07 \times 0,1$

$F_\mu = 131,947 \ \ kN$ (2)

[13]

QUESTION 2

2.1 $$M = Fl + \frac{w L^2}{2} = 3 \times 10^3 \times 2 + \frac{357,084 \times 5^2}{2} = 10,464 \quad kNm$$ (2)

2.2 $$\sigma_{top} = \frac{M Y_t}{I} = \frac{10,464 \times 10^3 \times 50 \times 10^{-3}}{10,662 \times 10^{-6}} = 49,069 \ \ MPa \quad (tensile)$$

$$\sigma_{bot} = \frac{M Y_c}{I} = \frac{10,464 \times 10^3 \times 100 \times 10^{-3}}{10,662 \times 10^{-6}} = 98,139 \ \ MPa \ \ (compressive)$$ (4)

2.3 $$y_1 = \frac{F l^3}{3 EI} = \frac{3 \times 10^3 \times 2^3}{3 \times 200 \times 10^9 \times 10,662 \times 10^{-6}} = 3,752 \ \ mm$$

$$y_2 = \frac{F l^2}{2 EI} \times l_f = \frac{3 \times 10^3 \times 2^2}{2 \times 200 \times 10^9 \times 10,662 \times 10^{-6}} \times 3 \ = 8,441 \ \ mm$$

$$y_3 = \frac{w L^4}{8 EI} = \frac{357,084 \times 5^4}{8 \times 200 \times 10^9 \times 10,662 \times 10^{-6}} = 13,083 \ \ mm$$

$y_T = y_1 + y_2 + y_3 = 25,275 \ \ mm$ (5)

[11]

QUESTION 3

3.1 $A = B \times D = B \times 3B = 3B^2$

$$\sigma_d = \frac{F}{A} = \frac{210 \times 10^3}{3B^2} = \frac{70 \times 10^3}{B^2}$$

$$I = \frac{DB^3}{12} = \frac{3B \times B^3}{12} = 0{,}25B^4$$

$$\sigma_b = \frac{FeY}{I} = \frac{210 \times 10^3 \times 0{,}25B \times 0{,}5B}{0{,}25B^4} = \frac{105 \times 10^3}{B^2}$$

$$\sigma_d + \sigma_b = \sigma_{max}$$

$$\frac{70 \times 10^3}{B^2} + \frac{105 \times 10^3}{B^2} = 100 \times 10^6$$

$$B = 41{,}833 \ mm$$
$$D = 125{,}499 \ mm$$

(7)

3.2 $$\sigma_d = \frac{70 \times 10^3}{B^2} = \frac{70 \times 10^3}{(41{,}833 \times 10^{-3})^2} = 40 \ MPa \ (tensile)$$

$$\sigma_b = \frac{105 \times 10^3}{B^2} = \frac{62{,}5 \times 10^3}{(41{,}833 \times 10^{-3})^2} = 60 \ MPa \ (tensile \vee compressive)$$

(2)

3.3 $\sigma_{min} = \sigma_b - \sigma_d = 60 - 40 = 20 \ MPa \ (compressive)$ (2)

[11]

QUESTION 4

4.1 $W = \rho g A l = 2\ 400 \times 9{,}81 \times 0{,}5 \times B \times 6 \times 1 = 70\ 632B = V$

$$F_w = \frac{\rho g h^2}{2} = \frac{1\ 000 \times 9{,}81 \times 6^2}{2} = 176\ 580 \ N$$

Moments about the toe: $V \times x_R + F_w \times \frac{h}{3} = W \times x_1$

$$70\ 632B \times \frac{B}{3} + 176\ 580 \times \frac{6}{3} = 70\ 632B \times \frac{2B}{3}$$

$$B = 3{,}873 \ m$$

(6)

4.2 $W \sim M = W \times x_1 = 70\ \ 632 \times 3{,}873 \times 0{,}667 \times 3{,}873 = 706{,}32\ \ kNm$

$$F \sim M = F_w \times \frac{h}{3} = 176\ \ 580 \times 2 = 353{,}16\ \ kNm$$

$$FOS = \frac{W \sim M}{F \sim M} = \frac{706{,}32}{353{,}16} = 2\ \ (safe) \geq 2$$ (4)

4.3 $V = 70\ \ 632\, B = 70\ \ 632 \times 3{,}873 = 273{,}557\ \ kN$

$$FOS = \frac{\mu V}{F_w} = \frac{0{,}6 \times 273{,}557}{1\ \ 765{,}80} = 0{,}93\ \ (not\ \ safe)\ \ < 1{,}5$$ (3)

[13]

QUESTION 5

5.1 $$\frac{b\,n^2}{2} = m\,A_s(d - n)$$

$$\frac{0{,}5 \times n^2}{2} = 15 \times 942{,}478 \times 10^{-6}(0{,}6 - n)$$

$$250 \times 10^{-3}\ \ n^2 = 8{,}482 \times 10^{-3} - 14{,}137 \times 10^{-3}\ \ n$$

$$250 \times 10^{-3}\ \ n^2 + 14{,}137 \times 10^{-3}\ \ n - 8{,}482 \times 10^{-3} = 0$$

$$n = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-14{,}137 \times 10^{-3} \pm \sqrt{(14{,}137 \times 10^{-3})^2 - 4(250 \times 10^{-3})(-8{,}482 \times 10^{-3})}}{2 \times 250 \times 10^{-3}}$$

$n = 158{,}082\ \ mm$ (4)

5.2 $M = \sigma_s A_s l_a = 110 \times 10^6 \times 942{,}478 \times 10^{-6} \times 0{,}547 = 56{,}741\ \ kNm$ (2)

5.3 $M = 0{,}5\,\sigma_c A_c l_a = 0{,}5 \times 3 \times 10^6 \times 0{,}5 \times 0{,}158 \times 0{,}547 = 64{,}889\ \ kNm$ (2)

5.4 $M = 56{,}741\ \ kNm\ \ (maximum\ \ allowed)$ (1)

5.5 $$M = 0{,}5\,\sigma_c A_c l_a$$

$$56{,}741 \times 10^3 = 0{,}5 \times \sigma_c \times 0{,}5 \times 0{,}158 \times 0{,}547$$

$$\sigma_c = 2{,}623\ \ MPa$$ (2)

5.6 $$w = \frac{2M}{L^2} = \frac{2 \times 56{,}741 \times 10^3}{5^2} = 4{,}539\ \ kN/m$$ (2)

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[13]

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QUESTION 6

6.1 $$y_B = \frac{F_{TB}}{w} = \frac{3\ 000}{30} = 100\ \ m$$

$$y_A = y_B - h = 100 - 5 = 95\ \ m$$

$$F_{TA} = w\,y_A = 30 \times 95 = 2\ \ 850\ \ N$$ (3)

6.2 $$y_0 = y_A - d = 95 - 2 = 93\ \ m$$

$$l_A = \sqrt{y_A^2 - y_0^2} = \sqrt{95^2 - 93^2} = 19{,}391\ \ m$$

$$l_B = \sqrt{y_B^2 - y_0^2} = \sqrt{100^2 - 93^2} = 36{,}756\ \ m$$

$$l_T = l_A + l_B = 56{,}147\ \ m$$

(4)

6.3 $$F_{vc} = w\,l_B = 30 \times 36{,}756 = 1\ \ 102{,}679\ \ N$$

$$F_{va} = F_T \cos\alpha = 3\ \ 000 \times \cos 60 = 1\ \ 500\ \ N$$

$$R_v = F_{va} + F_{vc} = 2\ \ 602{,}679\ \ N$$ (3)

6.4 $$F_{hc} = w\,y_0 = 30 \times 93 = 2\ \ 790\ \ N$$

$$F_{ha} = F_T \sin\alpha = 3\ \ 000 \times \sin 60 = 2\ \ 598{,}076\ \ N$$

$$R_h = F_{hc} - F_{ha} = 191{,}924\ \ N$$

$$R = \sqrt{2\ \ 602{,}679^2 + 191{,}924^2} = 2\ \ 609{,}745\ \ N$$ (4)

[14]

QUESTION 7

7.1 $$T_e = \frac{\pi d^3 \tau}{16} = \frac{\pi \times 0{,}08^3 \times 50 \times 10^6}{16} = 5{,}027 \ \ kNm$$

$$M^2 + T^2 = T_e^2$$

$$M^2 + 4^2 = 5{,}027^2$$

$$M = 3{,}044 \ \ kNm \qquad (3)$$

7.2 $$M_e = \frac{\pi d^3 \sigma}{32} = \frac{\pi \times 0{,}08^3 \times 75 \times 10^6}{32} = 3{,}77 \ \ kNm$$

$$M_e = 0{,}5\left[M + \sqrt{M^2 + T^2}\right]$$

$$3{,}77 = 0{,}5\left[M + \sqrt{M^2 + 4^2}\right]$$

$$7{,}54 - M = \sqrt{M^2 + 4^2}$$

$$(7{,}54 - M)^2 = M^2 + 4^2$$

$$7{,}54^2 - 15{,}08 \ \ M + M^2 = M^2 + 4^2$$

$$M = 2{,}709 \ \ kNm \qquad (5)$$

7.3 The maximum BM allowed is 2,709 kNm.

REASON: This BM will satisfy both given stress limits. (2)

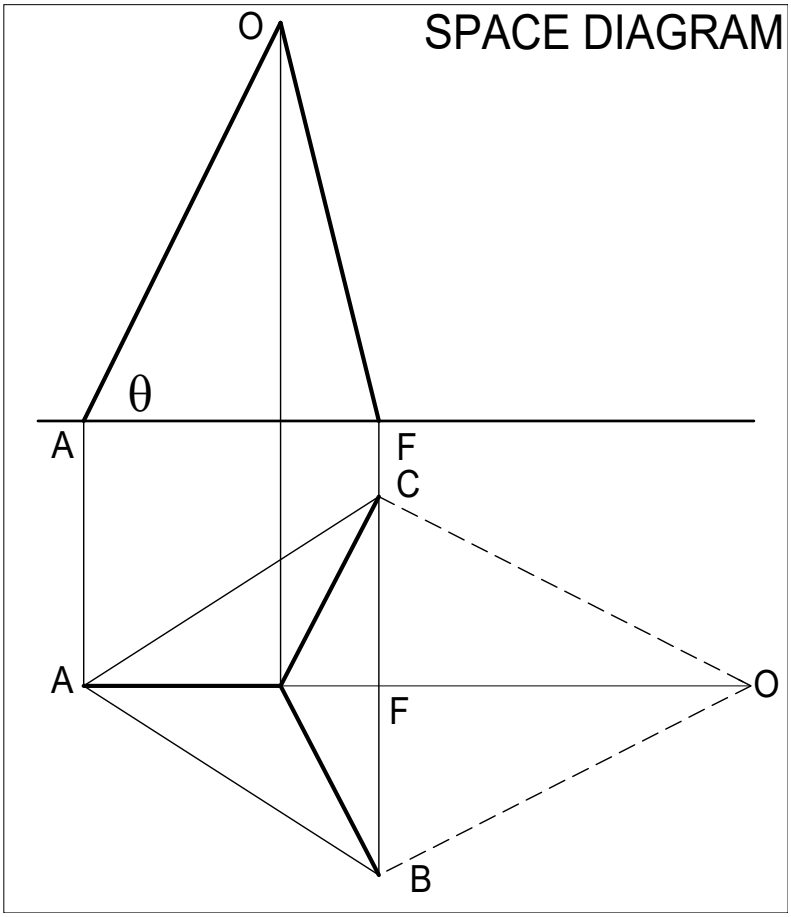
7.4 $$T_e = \sqrt{M^2 + T^2} = \sqrt{2{,}709^2 + 4^2} = 4{,}831 \ \ kNm$$

$$\tau = \frac{16\,T_e}{\pi d^3} = \frac{16 \times 4{,}831 \times 10^3}{\pi \times 0{,}08^3} = 48{,}054 \ \ MPa$$

(2)
[12]

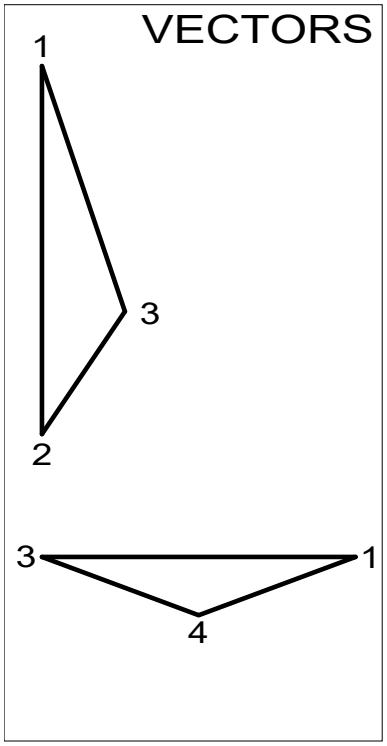
QUESTION 8

8.1



(4)

8.2



(4)

MEMBER	MAGNITUDE
OA (2→3)	2,28 kN✓
OC (3→4)	2,28 kN✓
OB (4→1)	2,28 kN✓

(3)

8.3 $$\mu = \frac{1}{\tan\theta} = \frac{1}{\tan 61} = 0{,}55$$

(2)

[13]

TOTAL: 100

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